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The Palaeomagnetism of Triassic and Cretaceous Rocks from Morocco

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Summary

Stable components of magnetization have been isolated in three formations of Triassic age (the Ait-Aadel dolerites, Issaldain dolerites and Titchka sandstone), and one formation of Cretaceous age (the Infracenomanian sandstone) in Central Morocco. When the new Triassic data are combined with existing African Triassic palaeomagnetic data an elongate distribution of palaeomagnetic poles can be identified for both north-west and south-east Africa which trend in a westerly direction away from the mean Mesozoic position. Possible interpretations of this distribution include Tertiary remagnetization, local tectonic rotation, motion of the African plate, and true polar wander, involving a short-lived westerly excursion of the pole. A similar, but apparently more-complicated distribution of poles is evident for the Cretaceous.

Introduction

Mesozoic palaeomagnetic data from north-west Africa are likely to be of particular use in investigations of the early history of evolution of the central North Atlantic basin, involving separation of eastern North America from north-western Africa. Furthermore, comparison of such data with Mesozoic palaeomagnetic data from south and east Africa would allow us to test the applicability of the geocentric dipole field model to Mesozoic time. For these reasons a palaeomagnetic study has been carried out on Mesozoic formations from the High Atlas and Anti-Atlas regions of Morocco. Results from Jurassic formations have already been published (Hailwood & Mitchell 1971) and this paper describes results from Triassic and Cretaceous formations.

Geology

(i) *Triassic*

The sequence of Triassic deposits in Morocco can be subdivided into two main lithological groups (Choubert & Faure-Muret 1962), comprising a Lower Series of red sandstones resting unconformably on Lower Permian or older deposits, and an Upper Series of red clays, often saliferous, and including towards the centre of the sequence a thick series of basaltic lava flows. An important marine incursion occurred during the eruption of the basalts, and is marked by a limestone intercalation containing a fauna characteristic of the German Middle Triassic Letterkhole (Upper Muschelkalk) stage. The distribution of the Lower Series of red sandstones is relatively

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limited and they are confined mainly to localized basins of subsidence. The deposits of the Upper Series are more widely distributed, and occur over most of the Atlas region. They indicate a widespread area of lagunal and lacustrine sedimentation extending over most of central Morocco during Mid-Triassic times. The source of sediments appears to have been the Hercynian chains lying to the south, since the sandy detrital deposits of the Lower Series are best developed in the sourthern part of the region. The volcanic phase which occurred during the deposition of the Upper Series is represented by a thick sequence of tholeiitic dolerites and basalts, reaching a maximum thickness of some 200 m. in the Central High Atlas. A series of continuous flows is interstratified with clays, formed from the erosion of the basalts, and baked red mudstones. The volcanic series has been highly altered and it seems probable that this occurred during or soon after the emplacement of the rocks, as a direct results of extrusion into an aqueous environment. If this is so, then the primary component of magnetic remanence carried by the alteration products should reflect the ambient geomagnetic field direction relative to this region during Middle Triassic times. Evidence for emplacement in a subaqueous environment includes the presence of interstratified clays, the existence in the basalts of very small grains of minerals in the form of 'spherulites' characteristic of rapid crystallization, the absence of flow structures, and the frequent presence, especially in fractured zones, of ferriferous jaspers which formed during the eruption of the lavas from water highly charged with silica (Broussé 1966). In addition, the whole series is characterized by an abundance of vacuoles, filled with chalcedony and chlorite. The alteration is most pronounced in the fracture zones and along faults, probably due to the easy penetration of water along the numerous joints and fissures, formed during rapid cooling and contraction of the lava.

The Moroccan Triassic formations were sampled at two localities in the doleritic basalts (Issaldain and Ait-Aadel), and at one locality in the Upper Series of red sandstones (Tizi n Titchka pass); the palaeomagnetic results will be discussed with reference to these locality names.

(ii) *Cretaceous*

Most of the important present-day structural features of south and central Morocco appear to have developed during Upper Cretaceous and Lower Tertiary times, and were only slightly modified during the ensuing Alpine orogenic movements. During the Lower Cretaceous (Infracenomanian) widespread continental deposits of Saharan origin were layed down and, at the beginning of the Cenomanian, a very important marine transgression resulted in the deposition of a sequence of red clays and highly fossiliferous limestones. Palaeomagnetic results from the Infracenomanian sediments are presented in this paper.

Palaeomagnetic results from Triassic formations

(i) *Issaldain dolerites*

A total of eight sites (MB28 to 30 and MB32 to 36) were sampled at this locality, on the southern flank of the Central High Atlas mountains, near the village of Issaldain on the Marrakech–Ouarzazate road (Fig. 1). As far as possible six separately oriented samples were collected at each site, and individual sites were located in separate lava flows.

Between-site and within-site scatter of natural remanent magnetization (NRM) directions is small (Table 1, and Fig. 2(a)), and the overall mean direction ($D = 3{\cdot}5°$, $I = 42°$) is close to the present geomagnetic field direction at these localities ($D = 351{\cdot}5°$, $I = 47{\cdot}5°$). Four pilot samples, selected at random from the eight sites, showed a high degree of stability to alternating magnetic field (AF) demagnetization,

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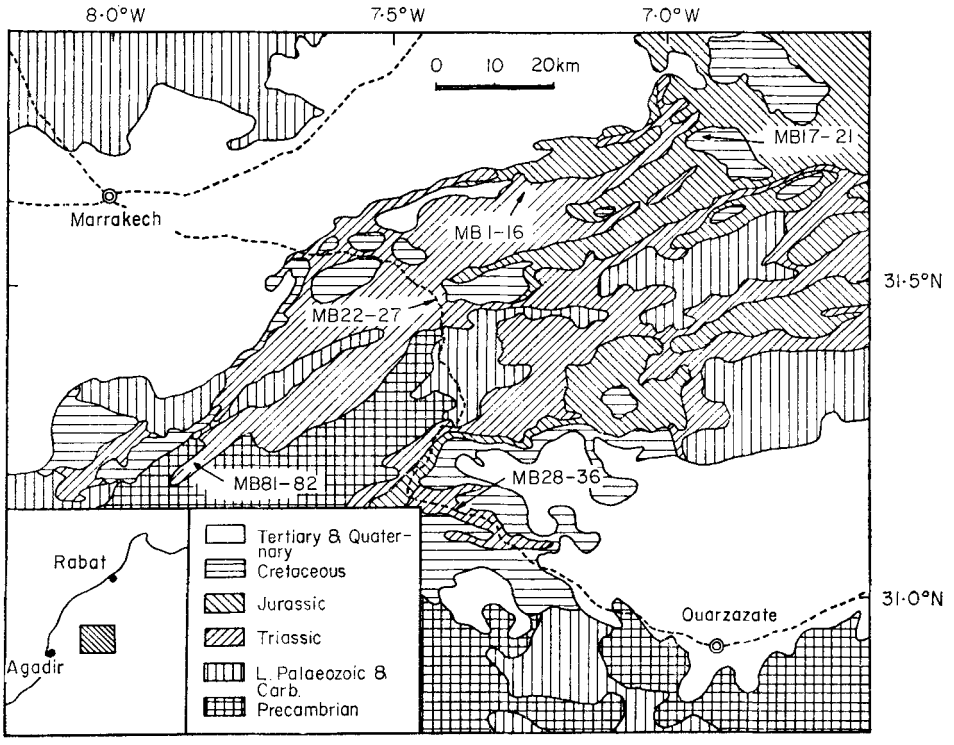


FIG. 1 Outline geology and location of sampling sites.

and generally small changes in direction of remanence towards the north or north-east (Fig. 2(b) and (c)). The temporary movement away from the main group of directions shown by sample MB29·2 at 750 oe is probably due to a measurement error, and the behaviour of sample MB35·1 above 300 oe may be due to the acquisition of a spurious anhysteretic component. After bulk demagnetization of the remaining samples from this locality, at fields chosen on the basis of the pilot sample behaviour, the mean site directions move away from the present geomagnetic field direction, and the between-site scatter remains unchanged (Table 1). This is interpreted as being due to the successful removal of a weak viscous component.

(ii) *Ait-Aadel dolerites*

AF demagnetization of one pilot sample from each site resulted in complex behaviour, probably associated with the introduction of anhysteretic components due to imperfect experimental conditions (Hailwood & Molyneux, 1974). Pilot samples from five sites were therefore subjected to progressive thermal demagnetization at 100 °C intervals up to 600 °C, and the results (Fig. 3(b) and (c)) indicate the isolation of a stable component with a NNW declination and intermediate inclination, on treatment at 100–500 °C. Bulk demagnetization of the remaining samples at 300 °C led to a substantial reduction of between-site scatter, from 24° to 7·5° (Table 2). One site (MB4) was eliminated from the final statistical analysis because of its excessively high within-site scatter (>90°).

(iii) *Titchka sediments*

A sequence of red sandstones lying in the detrital Upper Series immediately below the Middle Triassic volcanic sequence was sampled at six sites situated near the Tizi

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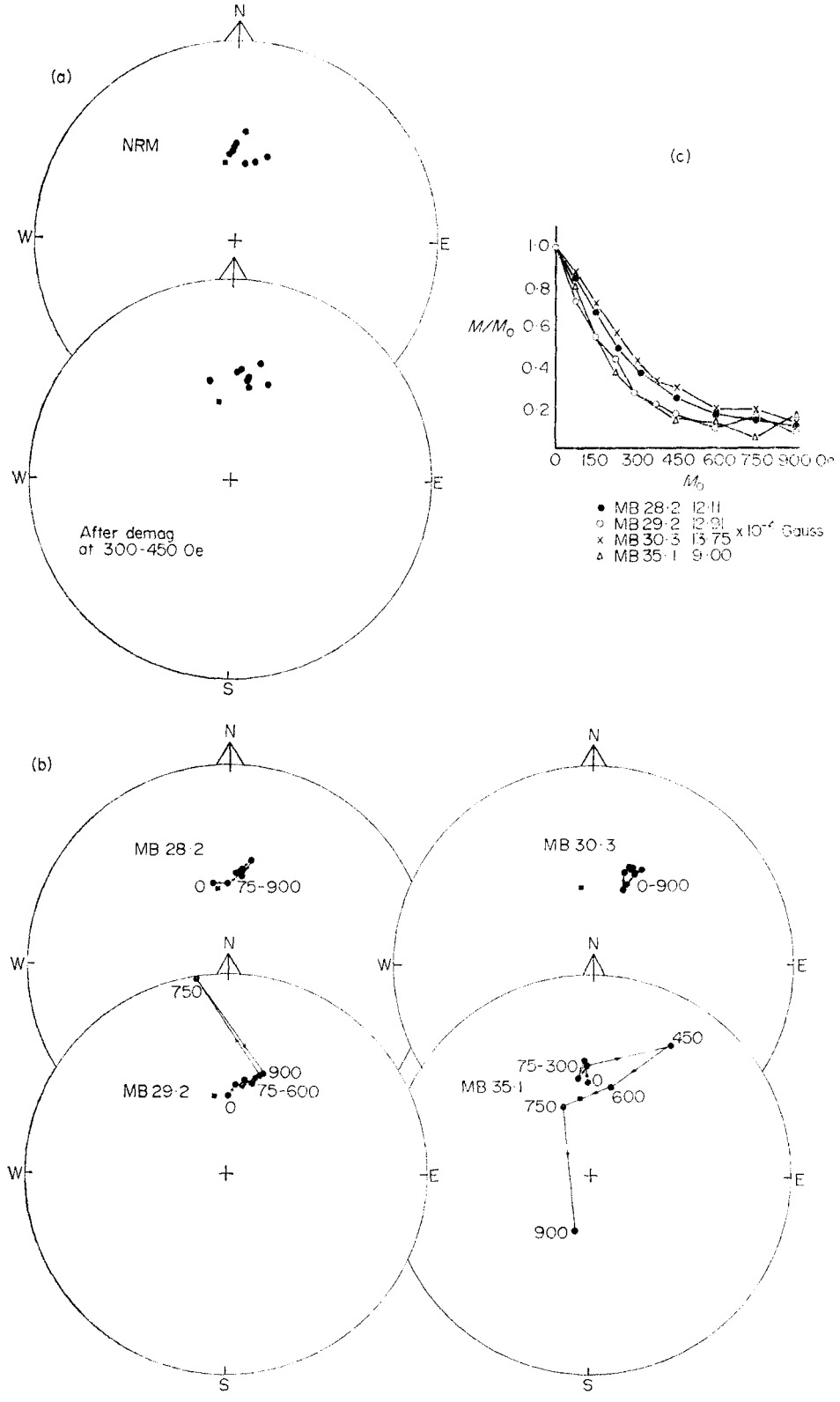


FIG. 2. Issaldain dolerites. (a) Mean site directions of remanence before and after AF demagnetization at 300–450 Oe; (b) Changes in direction of remanence; and (c) changes in intensity of magnetization of pilot samples during progressive AF demagnetization. Solid symbols lower hemisphere, open symbols upper hemisphere. Numbers indicate peak field in Oersteds. Black square indicates present field direction. Stereographic projection.

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Table 1

Issaldain Triassic dolerites. Mean site directions, pole positions and statistical parameters

Site No.	NRM						After bulk AF demag.						Pole position		
	D	I	N	R	k	α_{95}	D	I	N	R	k	α_{95}	Lat.	Long.	A_{95}
MB28	356·9	+41·8	5	4·896	38	12	9·2	+37·2	6	5·727	18	16	77·7° N	130·0° E	14
MB29	12·7	+47·9	6	5·948	96	7	3·3	+33·4	6	5·865	37	11	77·1 N	153·8 E	10
MB30	18·9	+42·5	6	5·991	564	3	20·6	+35·6	6	5·979	234	4	68·2 N	109·2 E	5
MB32	4·1	+32·8	5	4·840	25	16	10·2	+34·8	5	4·849	26	15	75·4 N	134·5 E	16
MB33	359·0	+38·5	6	5·982	279	4	14·3	+27·9	6	5·965	145	6	69·2 N	130·6 E	4
MB34	357·6	+40·4	6	5·996	1162	2	5·2	+32·0	6	5·981	258	4	75·5 N	152·4 E	4
MB35	354·8	+42·6	6	5·985	334	4	347·9	+36·3	5	4·929	57	10	74·6 N	221·3 E	10
MB36	6·1	+48·1	6	5·927	69	8	10·8	+40·0	5	4·273	5	36	82·8 N	111·5 E	33
Overall mean	3·6	+42·1	8	7·933	104	6	7·8	+35·0	8	7·921	88	6	77·3 N	140·6 E	6·5

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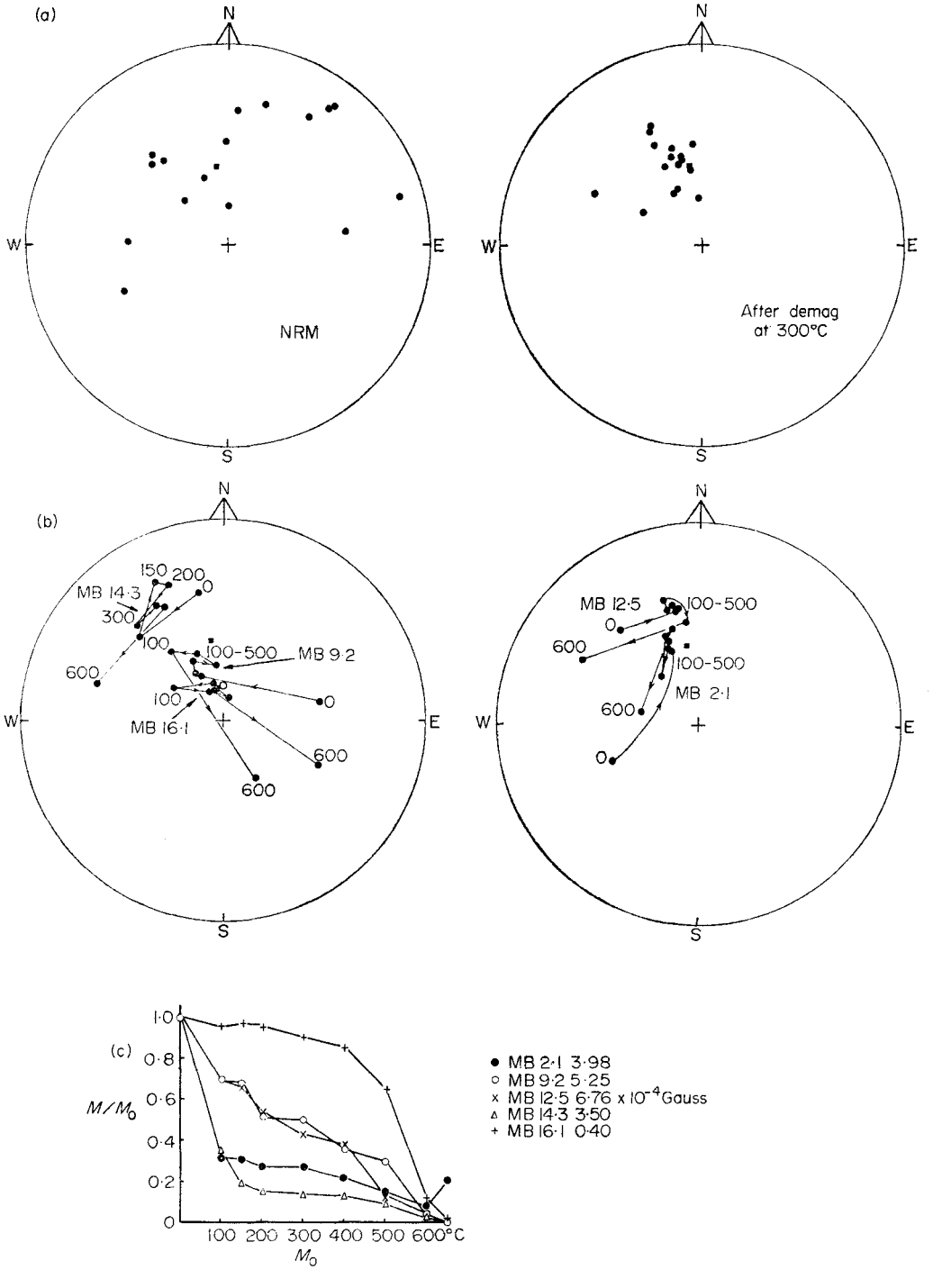


FIG. 3. Ait-Aadel dolerites. (a) Mean site directions of remanence before and after thermal demagnetization at 300 °C; (b) changes in direction of remanence; and (c) changes in intensity of magnetization of pilot samples during progressive thermal demagnetization (symbols as in Fig. 2).

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Table 2

Ait-Aadel Triassic dolerites. Mean site directions, pole positions and statistical parameters

Site	NRM						After thermal demag. at 300 °C						Pole Position		
No.	D	I	N	R	k	α_{95}	D	I	N	R	k	α_{95}	Lat.	Long.	A_{95}
MB1	14·6	+18·9	6	5·706	17	17	341·3	+41·0	4	3·904	31	17	72·2° N	115·9° W	15
MB2	245·9	+31·9	6	5·220	6	29	344·5	+45·2	4	3·977	129	8	75·7 N	112·2 W	7
MB3	315·9	+55·6	5	4·670	12	33	336·8	+24·6	4	2·865	3	71	66·5 N	124·6 W	58
MB4	316·9	+32·7	5	4·934	61	10	300·1	+54·0	5	2·443	2	GT90	22·5 N	62·0 W	GT90
MB5	4·0	+22·8	5	4·758	16	19	337·4	+51·5	5	3·686	3	53	66·9 N	69·5 W	71
MB6	73·8	+8·0	6	4·895	4	35	296·5	+30·2	6	4·905	4	35	32·3 N	93·9 W	34
MB7	32·5	+17·4	6	4·639	4	41	354·6	+37·0	4	3·218	4	54	80·1 N	124·1 W	62
MB8	36·0	+10·4	5	4·917	48	11	332·0	+58·1	4	3·134	4	58	70·5 N	66·5 W	82
MB9	83·0	+29·6	6	5·675	15	18	357·3	+41·7	3	2·976	83	14	82·2 N	166·8 W	12
MB10	37·5	+9·0	6	5·858	35	11	343·0	+37·4	4	3·927	41	14	71·3 N	126·1 W	12
MB11	271·9	+38·5	6	5·660	15	18	335·4	+32·9	4	3·992	371	5	63·9 N	123·0 W	3
MB12	320·4	+30·0	3	2·996	556	5	346·7	+44·0	1	1·000	—	—	76·9 N	120·4 W	—
MB13	339·7	+51·5	5	4·957	94	8	351·9	+49·0	4	3·958	71	11	83·2 N	103·9 W	11
MB14	358·6	+35·6	4	3·510	6	40	336·0	+27·2	2	1·963	27	50	62·3 N	128·9 W	32
MB15	323·0	+35·4	6	5·410	8	24	335·5	+43·6	4	3·963	82	10	67·8 N	106·6 W	11
MB16	0·0	+68·2	5	4·931	58	10	357·1	+64·2	5	4·917	48	11	74·0 N	17·8 W	15
Overall mean	0·0	+43·2	16	11·497	3	24	339·7	+42·8	15	14·482	27	7·5	72·0 N	105·5 W	7·5
												Antipole	72·0 N	74·5 E	7·5

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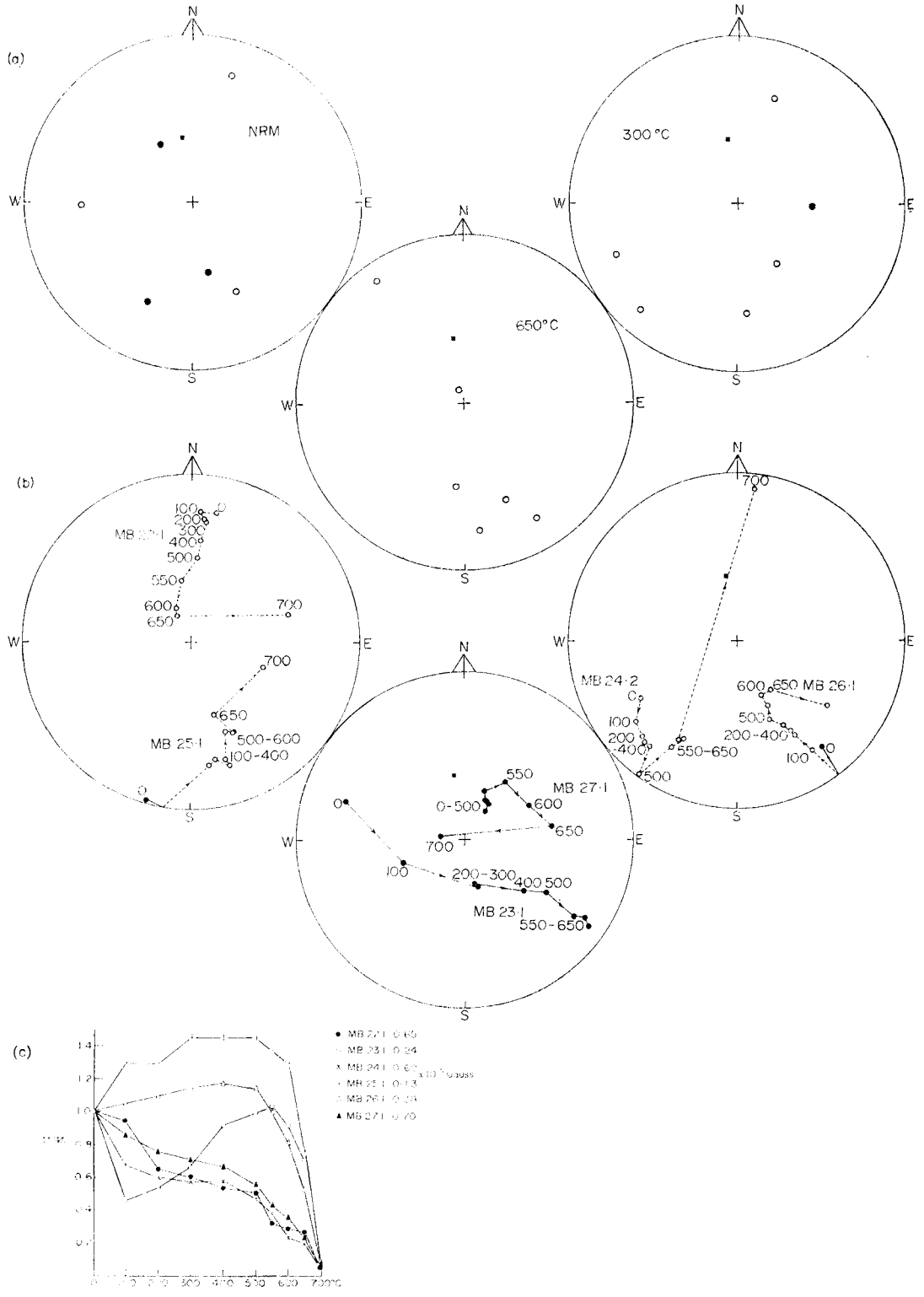


FIG. 4. Titchka sediments. (a) Mean site directions of remanence before and after treatment at 300 °C and 650 °C; (b) changes in direction of remanence; and (c) changes in intensity of magnetization of pilot samples during progressive thermal demagnetization (symbols as in Fig. 2).

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Table 3

Titchka Triassic sediments. Mean site directions, pole positions and statistical parameters

Site	NRM			300 °C			650 °C			Pole position		
No.	D	I	α_{95}	D	I	α_{95}	D	I	α_{95}	Lat.	Long.	A_{95}
MB22	17·0	−13·1	13·5	19·4	−22·6	40·4	339·9	−80·4	38·8	16·9° N	172·1° E	52·3
MB23	331·5	+46·7	33·1	92·5	+42·2	83·7	148·5	−11·9	35·8	51·9 S	69·0 E	34·7
MB24	268·8	−22·8	38·4	247·5	−14·3	38·7	185·7	−37·0	38·8	69·9 S	141·2 E	53·3
MB25	203·8	+24·3	33·2	175·0	−23·3	30·3	172·8	−15·4	18·7	73·8 S	30·8 E	17·4
MB26	151·7	−29·4	48·2	147·1	−43·9	34·1	156·8	−26·1	23·3	61·1 S	103·7 E	31·2
MB27	167·5	+43·8	65·8	222·5	−9·2	GT90	325·1	−6·4	82·0	53·2 N	237·0 E	80·1
Overall mean	223·9	+33·4	GT90	181·4	−34·2	GT90	173·5	−46·7	63·5	57·5 S	139·0 E	81·9
				Mean of sites MB23–26 only			165·1	−23·2	21·8	68·5 S	86·8 E	22·5

$N = 6$ cores per site for all six sites.

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n Titchka pass on the Marrakech–Ouarzazate road (sites MB22 to 27). The intensities of NRM of the sediments are weak, and lie in the range $0{\cdot}1$ to $0{\cdot}7 \times 10^{-5}$ G. The NRM mean site directions are plotted in Fig. 4, and listed in Table 3. With the exception of site MB22, the within-site scatter of directions is high ($>30°$), and the between-site scatter is greater than 90°. On progressive thermal demagnetization at 50 or 100 °C intervals up to 650 °C the remanence vectors of pilot samples from all sites moves towards a negative south to south-easterly direction (Fig. 4(b)), and in some cases the intensity decay curves indicate the possible presence of two approximately oppositely aligned components (e.g. samples MB23·1 and 25·1, Fig. 4(c)). The observed directional changes suggest the presence of a stable underlying component with a negative south to south-easterly direction, and in an attempt to isolate this component the remaining samples from all six sites were demagnetized at both 300 °C and 650 °C. Treatment at 300 °C does not significantly reduce the between-site scatter, and the within-site scatter is increased in four out of the six sites (Table 3 and Fig. 4(a)). However treatment at 650 °C results in movement of the mean direction of remanence of four of the six sites to the negative south-easterly quadrant. Of the other two sites, MB27 retains an excessively high within-site scatter ($\alpha_{95} = 82°$) and MB22 does not appear to have reached the terminal direction indicated by the other four sites.

It is considered that a common stable component of remanence has been isolated in the four sites MB23 to 26 after treatment at 650 °C, and this high coercivity component is probably carried by ilmeno-haematite. The mean direction of magnetization and mean palaeomagnetic pole-position are listed in Table 3. Although the between-site scatter for these four sites is reduced after application of a tilt correction based on the dip of the bedding planes, the difference is not significant at the 95 per cent confidence level (McElhinny 1964), so that the 'fold' stability test remains inconclusive in this case.

Six samples were collected from a further two sites in the Triassic red sandstone series, near the summit of Jebel Oukaimeden (sites MB81 and 82, Fig. 1), but exhibited predominantly unstable palaeomagnetic behaviour.

Palaeomagnetic results from Cretaceous sediments

The continental Infracenomanian deposits of the Central Atlas region were sampled at five sites (MB17 to 21), spanning a total stratigraphic thickness of some 100 m of coarse-grained red sandstones in the vicinity of Demnate. The intensities of NRM of these samples are generally higher than those of the Triassic red sediments, and vary from $0{\cdot}5$ to $1{\cdot}7 \times 10^{-5}$ G. This may possibly be correlated with their much darker red colouration. Both the between-site and the within-site scatter of NRM directions is small (Table 4). Small bedding corrections of up to 8° were applied to the remanence vectors, but had no significant effect on the between-site scatter. The NRM mean site directions all lie just to the NW of the present geomagnetic field direction at the sampling locality (Fig. 5(a)) and the latter lies within the 95 per cent confidence limit of the overall mean direction. One sample from each site was subsubjected to progressive thermal demagnetization up to a maximum temperature of 675 °C (Figs 5(b) and (c).) In all cases the direction of remanence undergoes little change on treatment up to 600 °C, after which an apparently random scatter of directions is observed. Although the NRM within-site and between-site scatter is small, all remaining samples were demagnetised at 300 °C to see if any significant changes occurred. The mean direction of remanence after treatment is identical to that before treatment, but there is a slight increase of between-site scatter (from 6° to 7°) and a general small increase of within-site scatter (Table 4), perhaps associated with the difficulties of measuring the weaker intensities. For this reason the mean pole-position was calculated from the results before demagnetization.

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Table 4

Infracenomanian sediments. Mean site directions, pole positions and statistical parameters

	Site No.	D	I	N	R	k	α_{95}		Lat.	Long.	A_{95}
(a) NRM											
	MB17	346·5	+34·0	6	5·882	42	10·5		72·3° N	140·0° W	11
	MB18	342·5	+35·0	6	5·926	67	8		69·9 N	130·0 W	8·5
	MB19	345·0	+45·5	5	4·919	49	11		75·9 N	121·5 W	14·0
	MB20	348·0	+39·5	6	5·971	174	5		75·9 N	133·5 W	4·5
	MB21	355·5	+40·0	5	4·964	113	7		80·5 N	162·5 W	6·5
Overall mean		347·5	+39·0	5	4·978	183	6·0		75·5 N	143·5 W	5·5
								Antipole	75·5 S	36·5 E	
(b) After thermal demagnetization at 300 °C											
	MB17	343·2	+29·6	6	5·878	41	10·5		68·6 N	137·5 W	9·5
	MB18	345·5	+33·5	6	5·922	64	8·5		71·5 N	137·0 W	8·5
	MB19	346·4	+47·9	5	4·887	35	13		77·7 N	103·0 W	16·5
	MB20	349·8	+38·8	6	5·965	141	5·5		77·1 N	139·5 W	5·0
	MB21	350·8	+41·5	5	4·953	84	8·5		79·5 N	136·0 W	8·0
Overall mean		347·0	+38·5	5	4·966	117	7·0		75·0 N	133·0 W	5·5
								Antipole	75·0 S	47·0 E	

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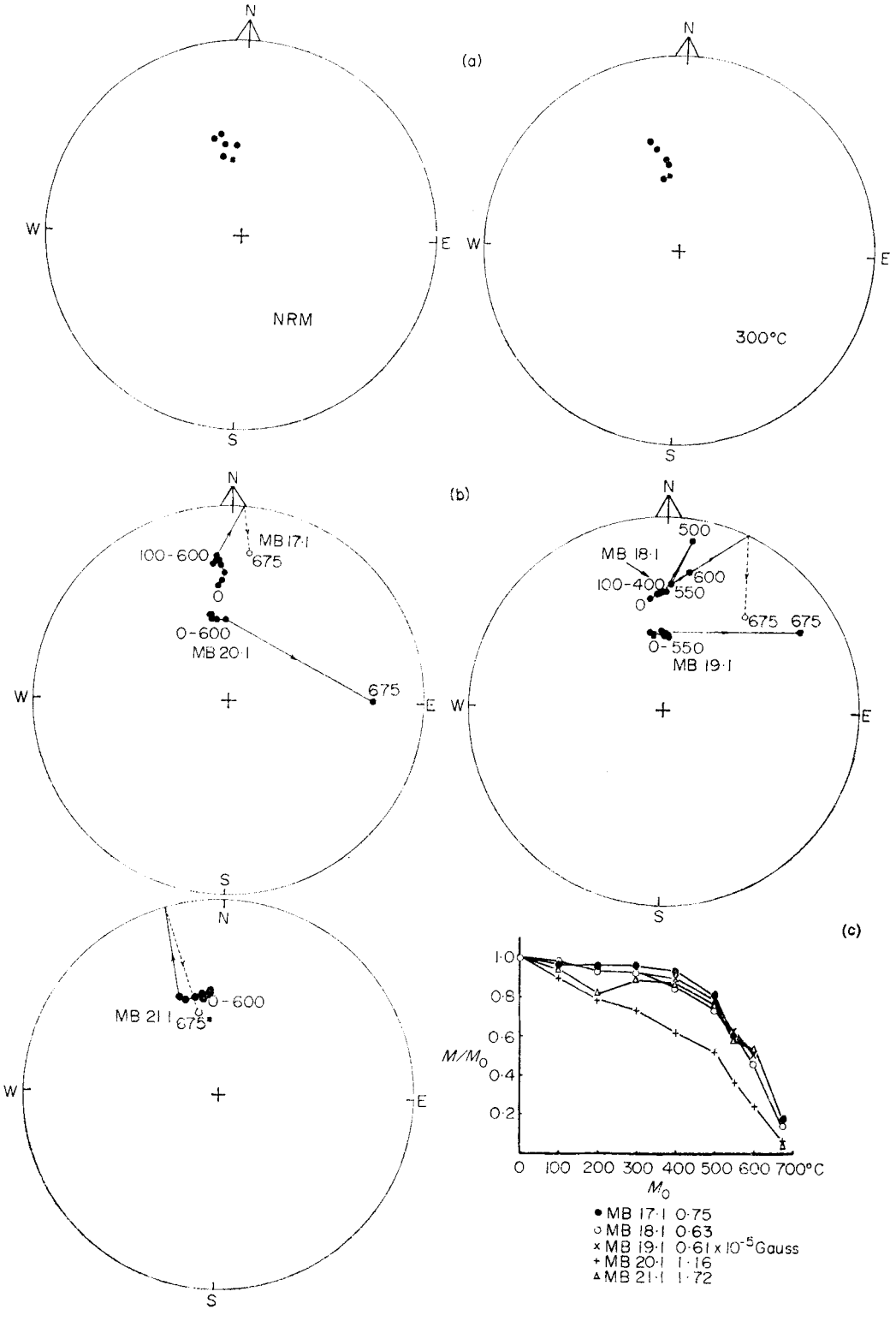


FIG. 5. Infracenomanian sediments. (a) Mean site directions of remanence before and after treatment at 300 °C; (b) changes in direction of remanence; and (c) changes in intensity of magnetization of pilot samples during progressive thermal demagnetization (symbols as in Fig. 2).

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Table 5

Summary of African Triassic and Cretaceous pole-positions

Pole Ref. No (Fig. 6)	Formation	Mean direction D	I	α_{95}	No. samples	No. sites	Mean pole position Lat	Long	A_{95}	Reference
(i) NW African Triassic poles (Fig. 6(a))										
1	Issaldain dolerites, Morocco	8·0	+35·0	6	45	8	77·5° S	39·5° W	6·5	This paper
2	Ait Aadel dolerites, Morocco	339·5	+43·0	7·5	63	15	72·0 S	74·5 E	7·5	This paper
3	Titchka sediments, Morocco	165·0	−23·0	22	22	4	68·5 S	87·0 E	22·5	This paper
4	Moroccan volcanics	347·5	+31·5	7	160	27	71·0 S	36·0 E	7·0	Bardon *et al.* (1973a)
5	Freetown gabbro, Sierra Leone	352·0	+7·0	7	11	11	81·0 S	45·0 E	5·0	Briden *et al.* (1971)
(ii) SE African Triassic poles (Fig. 6(a))										
6	K5 red sandstones, Songwe Rift, Tanzania	14·0	−27·0	10	5	2	75·0 S	36·0 W	—	Nairn (1964)
7	Upper Beaufort sediments, S. Africa	—	—	—	19	4	67·0 S	87·0 E	—	Graham (1961)
8	Red sandstone formation, Zambia	350·5	−55·5	4·5	—	6	68·0 S	49·5 E	5·0	Opdyke (1964)
9	Shawa ijolite, Rhodesia	335·5	−51·5	11·5	13	5	64·0 S	85·5 E	14·0	Gough & Brock (1964)
(iii) Cretaceous poles (Fig. 6(b))										
1	Infracenomanian sandstones, Morocco	347·5	+39·0	6	28	5	75·0 S	36·5 E	5·5	This paper
2	Cretaceous volcanics Morocco	311·5	+32·0	10	78	11	44·0 S	71·0 E	10·0	Bardon *et al.* (1973b)
3	Lupata volcanic series, Mozambique	336·0	−54·5	3	—	7	62·0 S	79·5 E	4·0	Gough & Opdyke (1963)
4	Mlanje Massif syenite, Malawi	333·0	−54·0	9·5	8	1	60·0 S	82·0 E	12·0	Briden (1967)
5	Red siltstones, Mozambique	359·0	−46·0	6	5	1	79·0 S	28·0 E	—	Gough & Opdyke (1963)

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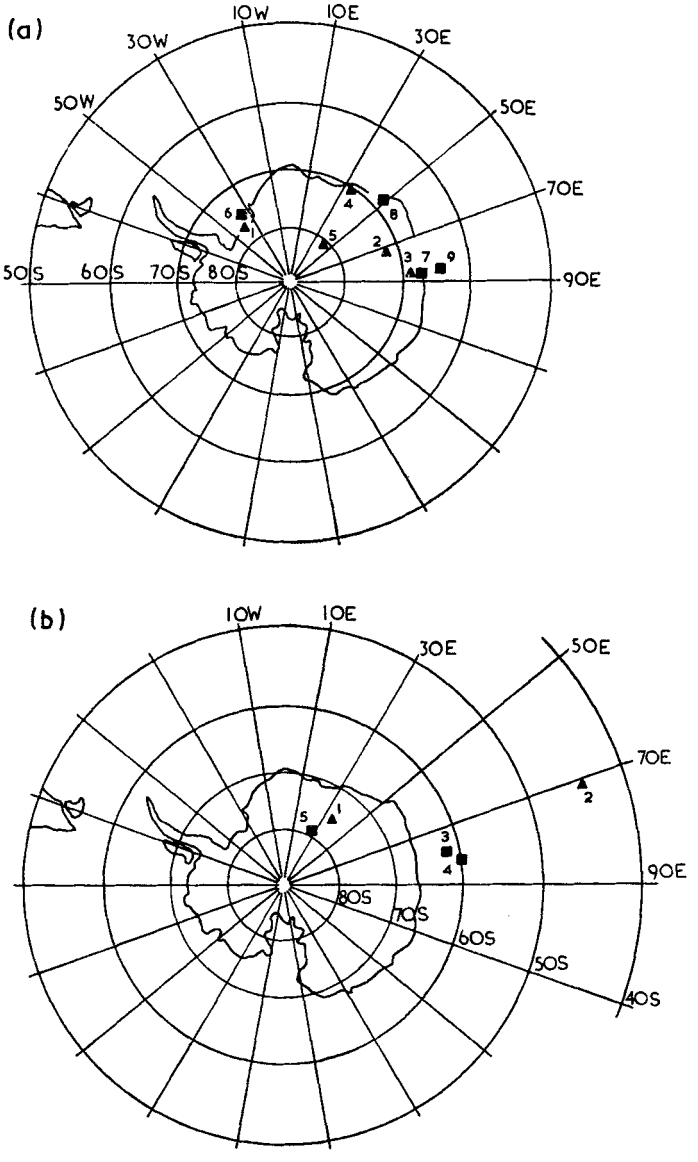


FIG. 6. African pole positions. (a) Triassic; and (b) Cretaceous. Small squares and triangles poles from SE and NW Africa, respectively. Pole numbers refer to Table 5.

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Discussion

(a) *Triassic*

The new Moroccan Triassic palaeomagnetic south pole positions described above are plotted in Fig. 6(a) (poles 1–3), together with all other African Triassic poles satisfying the two criteria:

(i) that they are based on AF or thermal stability tests; and

(ii) that the value of α_{95} in the final pole determination is $<25°$.

These poles exhibit an elongated distribution through some 110° of arc in a westerly direction away from the mean Mesozoic pole-position for Africa (approx. 65° S, 80° E). This distribution involves data from both NW Africa (triangles) and SE Africa (squares), and may be attributable to several possible factors:

1. *Tertiary remagnetization.* It is possible that the observed westerly stringing of poles represents the effects of varying degrees of remagnetization in Tertiary or Recent times, possibly as a result of chemical weathering. Poles 1 and 6 do in fact lie in the same general region as the mean Tertiary pole (Lat 85° S, Long 25° W), and although the present-day topography is undoubtedly very different from that of late Tertiary times, it is of interests to note that the Issaldain samples (pole 1) were collected from surface exposures, whereas the Ait Aadel samples (pole 2) were collected from deep underground tunnels.

2. *Tectonic rotation.* The westerly part of this distribution involves only one pole from SE Africa, that from the Tanzanian K5 Songwe Rift sandstones (pole 6, Fig. 6(a)). This pole is based on only 11 specimens from five samples at two sites, and must therefore be treated with reservation. If this pole is excluded, then the observed westerly distribution involves only NW African data, and could therefore represent the effects of relative tectonic movement between NW and SE Africa. However there seem to be no grounds for completely rejecting pole 6, since AF demagnetization tests indicate the presence of a stable component of remanence (Nairn 1964), and the low scatter of directions ($\alpha_{95} = 10°$) implies that measurement and orientation errors are not important. Furthermore the pole is derived from sediments which presumably did not acquire their remanence instantaneously, so that secular variation should be at least partially averaged out. In view of the remarkably good agreement between this pole and pole 1 from the other extremity of the African continent it is unlikely that the observed distribution is due to the effects of *local* tectonic rotation.

3. *Motion of the African plate:* The oldest pole contributing to the easterly part of the distribution shown in Fig. 6(a) is pole 7, of Lower Triassic age. Results from Jurassic and Cretaceous rocks (McElhinny *et al.* 1968; Hailwood & Mitchell 1971) indicate that the pole remained in the same approximate position relative to Africa until at least Lower Cretaceous times. Thus it is possible that the elongate distribution represents a short-lived episode of reversible motion of the African plate in Middle Triassic times.

4. *Polar wander:* An alternative possibility is that this distribution represents a temporary westward excursion of the pole relative to the stationary African plate. It should be possible to distinguish between the last two possibilities by searching for similar, synchronous westward excursions in the polar-wander curves from other continents (Hailwood & Tarling, in preparation).

(b) *Cretaceous*

Poles from African Cretaceous formations, satisfying the acceptance criteria stated above, are plotted in Fig. 6(b). Once again a westerly distribution of poles away from

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the mean Mesozoic position (represented by poles 3 and 4) is apparent, but is further complicated by the location of pole 2, from the Moroccan Lower Cretaceous volcanics (Bardon *et al.* 1973(b)).

The discrepancy between Moroccan poles 1 and 2 is difficult to explain, since both of these poles were derived from formations located within the same structural region of the High Atlas mountains, and are of closely similar age. The mean inclination of the sediments is 16° *higher* than that of the volcanics, and so is unlikely to reflect the effects of depositional inclination errors (e.g. Hospers 1967). The same conclusion applies to the discrepancy between the pair of poles 1 and 5, and the mean Mesozoic position, which is also mainly in declination. The close agreement between pole 1, from NW Africa, and pole 5, from SE Africa appears to be more than fortuitous, and suggests that this discrepancy is due to plate motion, or the effect of geographically extensive complexities of the geomagnetic field, rather than to relative tectonic motion.

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